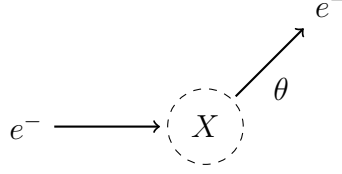


Rutherford scattering

Rutherford scattering is the interaction $e^- + X \rightarrow e^- + X$ where X is an atom.



These are the momentum vectors for Rutherford scattering where $E = \sqrt{p^2 + m^2}$.

$$p_1 = \begin{pmatrix} E \\ 0 \\ 0 \\ p \end{pmatrix} \quad p_2 = \begin{pmatrix} E \\ p \sin \theta \cos \phi \\ p \sin \theta \sin \phi \\ p \cos \theta \end{pmatrix}$$

inbound electron outbound electron

Spinors for the inbound electron.

$$u_{11} = \frac{1}{\sqrt{E+m}} \begin{pmatrix} E+m \\ 0 \\ p \\ 0 \end{pmatrix} \quad u_{12} = \frac{1}{\sqrt{E+m}} \begin{pmatrix} 0 \\ E+m \\ 0 \\ -p \end{pmatrix}$$

inbound electron spin up inbound electron spin down

Spinors for the outbound electron.

$$u_{21} = \frac{1}{\sqrt{E+m}} \begin{pmatrix} E+m \\ 0 \\ p_{2z} \\ p_{2x} + ip_{2y} \end{pmatrix} \quad u_{22} = \frac{1}{\sqrt{E+m}} \begin{pmatrix} 0 \\ E+m \\ p_{2x} - ip_{2y} \\ -p_{2z} \end{pmatrix}$$

outbound electron spin up outbound electron spin down

The probability density $|\mathcal{M}_{ab}|^2$ for spin state ab is

$$|\mathcal{M}_{ab}|^2 = \left(\frac{Z\alpha}{q^2} \right)^2 |\bar{u}_{2b} \gamma^0 u_{1a}|^2$$

where Z is atomic number of the nucleus and q^2 is momentum transfer such that

$$q^2 = (p_1 - p_2)^2 = (p_1 - p_2)^\mu g_{\mu\nu} (p_1 - p_2)^\nu = 2p^2(\cos \theta - 1)$$

The expected probability density $\langle |\mathcal{M}|^2 \rangle$ is the average of spin states.

$$\langle |\mathcal{M}|^2 \rangle = \frac{1}{2} \sum_{a=1}^2 \sum_{b=1}^2 |\mathcal{M}_{ab}|^2$$

The Casimir trick uses matrix arithmetic to sum over spin states.

$$\langle |\mathcal{M}|^2 \rangle = \frac{1}{2} \left(\frac{Z\alpha}{q^2} \right)^2 \text{Tr} \left[(\not{p}_1 + m) \gamma^0 (\not{p}_2 + m) \gamma^0 \right]$$

The result is

$$\langle |\mathcal{M}|^2 \rangle = 2 \left(\frac{Z\alpha}{q^2} \right)^2 (E^2 + m^2 + p^2 \cos \theta)$$

Recall that $m \approx 0.511 \text{ MeV}$. For $m \gg p$ we can use the approximation

$$E^2 + m^2 + p^2 \cos \theta \approx 2m^2$$

Hence

$$\langle |\mathcal{M}|^2 \rangle = \left(\frac{2mZ\alpha}{q^2} \right)^2$$

The differential cross section is equivalent to the expected probability density.

$$\frac{d\sigma}{d\Omega} = \langle |\mathcal{M}|^2 \rangle = \left(\frac{2mZ\alpha}{q^2} \right)^2$$

Substitute $q^2 = 2p^2(\cos \theta - 1)$ to obtain

$$\frac{d\sigma}{d\Omega} = \left[\frac{mZ\alpha}{p^2(\cos \theta - 1)} \right]^2$$

Noting that

$$p^2 = 2mE$$

we can also write

$$\frac{d\sigma}{d\Omega} = \left[\frac{Z\alpha}{2E(\cos \theta - 1)} \right]^2$$

Restore $\hbar$ and c to physical units.

$$\frac{d\sigma}{d\Omega} = \left[\frac{Z\alpha\hbar c}{2E(\cos \theta - 1)} \right]^2$$

Eigenmath script